Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 1 of 18 · Question 11 part a.i · 3 marks

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11 a i	$\left(\overrightarrow{AN}=\right)\overrightarrow{AO}+\overrightarrow{ON}$ or $\left(\overrightarrow{AN}=\right)\overrightarrow{AO}+\frac{3}{4}\overrightarrow{OB}=-\mathbf{a}+\frac{3}{4}\mathbf{b}$	M1 A1
ii	$\left(\overrightarrow{BM}=\overrightarrow{BO}+\overrightarrow{OM}=\right)-\mathbf{b}+\frac{1}{2}\mathbf{a}$	B1 (A1 on ePen) (3)
b	$\left(\overrightarrow{AX}=\lambda\left(\pm\left(\text{their }\overrightarrow{AN}\right)\right)\right)=\lambda\left(\pm"-\mathbf{a}+\frac{3}{4}\mathbf{b}"\right)$	B1 ft
	$\left(\overrightarrow{AX}=\overrightarrow{AM}+\mu\left(\pm\left(\text{their }\overrightarrow{MB}\right)\right)=\right)-\frac{1}{2}\mathbf{a}+\mu\left(\pm"-\frac{1}{2}\mathbf{a}+\mathbf{b}"\right)$	M1
	Equating components gives $-\lambda=-\frac{1}{2}-\frac{1}{2}\mu$	M1
	and $\frac{3}{4}\lambda=\mu$	M1
	$-\lambda=-\frac{1}{2}-\frac{1}{2}\left(\frac{3}{4}\lambda\right)\Rightarrow\lambda=\frac{4}{5}$ *or* $\mu=\frac{3}{5}$	ddM1 A1
	So $AX:XN=4:1$	A1 (7)
ALT	$\left(\overrightarrow{AX}=\overrightarrow{AB}+\overrightarrow{BN}+\lambda\left(\pm\left(\text{their }\overrightarrow{AN}\right)\right)\right)=\mathbf{b}-\mathbf{a}-\frac{1}{4}\mathbf{b}+\lambda\left(\pm"-\mathbf{a}+\frac{3}{4}\mathbf{b}"\right)$	B1ft
	$\left(\overrightarrow{AX}=\overrightarrow{AM}+\mu\left(\pm\left(\text{their }\overrightarrow{MB}\right)\right)=\right)-\frac{1}{2}\mathbf{a}+\mu\left(\pm"-\frac{1}{2}\mathbf{a}+\mathbf{b}"\right)$	M1
	Equating components gives $-\frac{1}{2}-\frac{1}{2}\mu=-1+\lambda$	M1
	and $\mu=\frac{3}{4}-\frac{3}{4}\lambda$	M1
	$\mu=\frac{3}{4}-\frac{3}{4}\left(\frac{1}{2}-\frac{1}{2}\mu\right)\Rightarrow\lambda=\frac{1}{5}$ *or* $\mu=\frac{3}{5}$	ddM1 A1
	So $AX:XN=4:1$	A1 (7)

General principles for marking part b

B1 ft states a valid vector, with a parameter, that could be used in a solution to find the ratio required.

M1 states a second valid vector path, with a second distinct parameter that could be used to find the ratio required. **This must be a distinct route, different to that used already, not travelling along the same line.**

M1 This mark is obtained by correctly equating their components for **a. They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ**

M1 This mark is obtained by correctly equating their components for **b. They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ**

ddM1 For an attempt to solve their simultaneous equations. There can be errors, but it must be clear they are solving simultaneous equations to arrive at a value for λ or μ or their parameters.

Dependent on both previous method marks.

A1 correct value for one of their parameters

A1 correct ratio

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Part	Mark	Additional Guidance
(a) (i)	M1	For a correct vector path stated, a fully correct answer implies this mark.
	A1	For $-\mathbf{a}+\frac{3}{4}\mathbf{b}$
(ii)	B1 (A1 ePen)	For $-\mathbf{b}+\frac{1}{2}\mathbf{a}$
(b)	B1 ft	For $\overrightarrow{AX}=\lambda\pm$ their $\overrightarrow{AN}$ written in terms of vectors **a** and **b** $\left(\overrightarrow{AX}=\right)\lambda\left(\pm\left("-\mathbf{a}+\frac{3}{4}\mathbf{b}"\right)\right)$ This may be seen embedded in working.
	M1	For e.g. $\overrightarrow{AX}=\overrightarrow{AM}+\mu\pm$ their $\overrightarrow{MB}$ written in terms of vectors **a** and **b** $\left(\overrightarrow{AX}=\right)-\frac{1}{2}\mathbf{a}+\mu\left(\pm\left("-\frac{1}{2}\mathbf{a}+\mathbf{b}"\right)\right)$ This may be seen embedded in working. **This must be a distinct route, different to that used already, not travelling along the same line.**
	M1	For $-\lambda=-\frac{1}{2}-\frac{1}{2}\mu$ This mark is obtained by correctly equating their components for **a**, the example given is when their equations are set up using $\overrightarrow{AX}=\lambda\overrightarrow{AN}$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\overrightarrow{MB}$ **They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ.**
	M1	For $\frac{3}{4}\lambda=\mu$ This mark is obtained by correctly equating their components for **b**, the example given is when their equations are set up using $\overrightarrow{AX}=\lambda\overrightarrow{AN}$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\overrightarrow{MB}$ **They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ.**
	ddM1	For an attempt to solve their simultaneous equations. There can be errors, but it must be clear they are solving simultaneous equations to arrive at a value for λ or μ or their parameters. Dependent on both previous method marks.
	A1	There are a number of different correct answers here, depending on how they've set up their equations: For $\lambda=\frac{4}{5}$ or $\mu=\frac{3}{5}$ using $\overrightarrow{AX}=\lambda\overrightarrow{AN}$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\overrightarrow{MB}$ For $\lambda=-\frac{4}{5}$ or $\mu=\frac{3}{5}$ using $\overrightarrow{AX}=\lambda\left(-\overrightarrow{AN}\right)$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\overrightarrow{MB}$ For $\lambda=\frac{4}{5}$ or $\mu=-\frac{3}{5}$ using $\overrightarrow{AX}=\lambda\overrightarrow{AN}$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\left(-\overrightarrow{MB}\right)$ For $\lambda=-\frac{4}{5}$ or $\mu=-\frac{3}{5}$ using $\overrightarrow{AX}=\lambda\left(-\overrightarrow{AN}\right)$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\left(-\overrightarrow{MB}\right)$
	A1	For $AX:XN=4:1$

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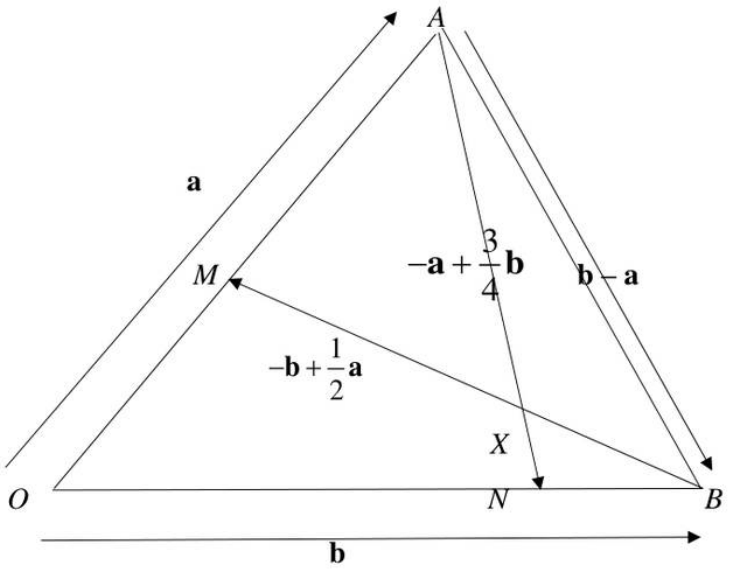
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(b) ALT	**B1ft**	For $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\left(\pm \text{their } \overrightarrow{AN}\right)$ written in terms of **a** and **b** $\left(\overrightarrow{AX}\right) = \mathbf{b} - \mathbf{a} - \frac{1}{4}\mathbf{b} + \lambda\left(\pm"-\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$ This may be seen embedded in working.
	M1	For e.g. $\overrightarrow{AX} = \overrightarrow{AM} + \mu \pm \text{their } \overrightarrow{MB}$ written in terms of vectors **a** and **b** $\left(\overrightarrow{AX} =\right) -\frac{1}{2}\mathbf{a} + \mu\left(\pm"-\frac{1}{2}\mathbf{a} + \mathbf{b}"\right)$ This may be seen embedded in working. **This must be a distinct route, different to that used already, not travelling along the same line.**
	M1	For $-\frac{1}{2} - \frac{1}{2}\mu = -1 + \lambda$ This mark is obtained by correctly equating their components for **a**, the example given is when their equations are set up using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\overrightarrow{AN}$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\overrightarrow{MB}$ **They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ.**
	M1	For $\mu = \frac{3}{4} - \frac{3}{4}\lambda$ This mark is obtained by correctly equating their components for **b**, the example given is when their equations are set up using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\overrightarrow{AN}$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\overrightarrow{MB}$ **They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ.**
	ddM1	For an attempt to solve their simultaneous equations. There can be errors, but it must be clear they are solving simultaneous equations to arrive at a value for λ or μ Dependent on both previous method marks.
	A1	There are a number of different correct answers here, depending on how they've set up their equations: For $\lambda = \frac{1}{5}$ or $\mu = \frac{3}{5}$ using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\overrightarrow{AN}$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\overrightarrow{MB}$ For $\lambda = -\frac{1}{5}$ or $\mu = \frac{3}{5}$ using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\left(-\overrightarrow{AN}\right)$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\overrightarrow{MB}$ For $\lambda = \frac{1}{5}$ or $\mu = -\frac{3}{5}$ using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\overrightarrow{AN}$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\left(-\overrightarrow{MB}\right)$ For $\lambda = -\frac{1}{5}$ or $\mu = -\frac{3}{5}$ using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\left(-\overrightarrow{AN}\right)$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\left(-\overrightarrow{MB}\right)$
	A1	For $AX : XN = 4 : 1$

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Answer 2 of 18 · Question 8 part a · 1 mark

2024 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

8(a)	$\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{DC} + \overrightarrow{CB} = 5\mathbf{a} - 2\mathbf{b} - 3\mathbf{a} = 2\mathbf{a} - 2\mathbf{b}$	B1 [1]
(b)	$\overrightarrow{BX} = k\overrightarrow{BD} = k(3\mathbf{a} + 2\mathbf{b})$ $\overrightarrow{BX} = \overrightarrow{BA} + \lambda\overrightarrow{AC} = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow k(3\mathbf{a} + 2\mathbf{b}) = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow 3k\mathbf{a} + 2k\mathbf{b} = (5\lambda - 2)\mathbf{a} + (2 - 2\lambda)\mathbf{b}$ $\Rightarrow 3k = 5\lambda - 2 \quad 2k = 2 - 2\lambda$ $\Rightarrow k = \frac{3}{8} \quad \left[\lambda = \frac{5}{8}\right]$	M1 M1 M1 M1A1 [5]
(c)	$\Delta CXD = \frac{5}{8}\Delta BCD$ $\Delta BCD = \frac{3}{8}ABCD$ $\frac{\Delta CXD}{ABCD} = \frac{\frac{5}{8}}{\frac{8}{3}} = \frac{15}{64}$ Ratio of area of triangle CXD : Trapezium $ABCD$ = 15 : 64	M1 M1 M1 A1 [4]
		Total 10 marks

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Answer 3 of 18 · Question 10 part a · 2 marks

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10 (a)	$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -(2\mathbf{a} - \mathbf{b}) + 3\mathbf{a} + \mathbf{b} = \mathbf{a} + 2\mathbf{b}$	M1A1 [2]
(b)	$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 3\mathbf{a} + \mathbf{b} - \mathbf{a} + 3\mathbf{b} = 2\mathbf{a} + 4\mathbf{b} = 2(\mathbf{a} + 2\mathbf{b})$ Conclusion required; same direction and $\overrightarrow{OC}$ is a multiple of $\overrightarrow{AB}$ therefore $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$.	M1 A1 [2]
(c)	$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \mathbf{a} + 2\mathbf{b} + (-\mathbf{a} + 3\mathbf{b}) = 5\mathbf{b}$ $\overrightarrow{AX} = \mu\overrightarrow{AC} = \mu 5\mathbf{b}$ $\overrightarrow{AX} = \overrightarrow{AO} + \overrightarrow{OX} = -(2\mathbf{a} - \mathbf{b}) + \lambda(3\mathbf{a} + \mathbf{b})$ $\Rightarrow \mu 5\mathbf{b} = -(2\mathbf{a} - \mathbf{b}) + \lambda(3\mathbf{a} + \mathbf{b})$ $\Rightarrow -2 + 3\lambda = 0 \Rightarrow \lambda = \frac{2}{3}$ $\Rightarrow 5\mu = 1 + \lambda \Rightarrow \mu = \frac{1}{3}$ $\Rightarrow AX : XC = 1 : 2$	B1 B1 M1 M1 M1 A1 A1 [7]
		Total 11 marks

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Question number	Scheme	Marks

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10 (a)	$\left(\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}\right) = 5\mathbf{p} + 3\mathbf{q}$	M1
	$\overrightarrow{DC} = \left(\overrightarrow{DO} + \overrightarrow{OC}\right) = \overrightarrow{DO} + \frac{3}{2}\overrightarrow{OB}$	M1
	$\overrightarrow{DC} = \frac{9}{2}(\mathbf{p} + \mathbf{q}) = \frac{9\mathbf{p}}{2} + \frac{9\mathbf{q}}{2}$	A1 (3)
(b)	$\overrightarrow{OF} = \overrightarrow{OD} + \lambda\overrightarrow{DC}$	M1
	$= 3\mathbf{p} + \frac{9}{2}\lambda(\mathbf{p} + \mathbf{q})$	A1
	$\overrightarrow{OF} = \overrightarrow{OA} + \mu\overrightarrow{AB}$	M1
	$= 5\mathbf{p} + 3\mu\mathbf{q}$	A1
	$3 + \frac{9}{2}\lambda = 5$	M1
	$\lambda = \frac{4}{9}$ $\left(\frac{9}{2}\lambda = 3\mu\right)$ or $\mu = \frac{2}{3}$	A1
	$\overrightarrow{OF} = 5\mathbf{p} + 2\mathbf{q}$	A1 (7)
(c)	$\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$	M1
	$\overrightarrow{OG} = 3\mathbf{p} + 2(\mathbf{p} + \mathbf{q}) + 5\beta\mathbf{p} = 5\mathbf{p} + 5\beta\mathbf{p} + 2\mathbf{q}$ ft their $\overrightarrow{OF}$	M1
	$3\alpha = 2 \Rightarrow \alpha = \frac{2}{3}$ $\left(\beta = -\frac{1}{3}\right)$	M1
	$\overrightarrow{OG} = \frac{10}{3}\mathbf{p} + 2\mathbf{q}$	A1 (4)
	ALT (c)	
	$\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$	{M1}
	$\overrightarrow{FG} = -5\mathbf{p} - 2\mathbf{q} + \alpha(5\mathbf{p} + 3\mathbf{q})$ ft their $-\left(\overrightarrow{OF}\right)$	{M1}
	As FG is parallel to AO there can be no $\mathbf{q}$ component in $\overrightarrow{FG}$	
	$3\alpha = 2 \Rightarrow \alpha = \frac{2}{3}$	{M1}
	$\overrightarrow{OG} = \frac{10}{3}\mathbf{p} + 2\mathbf{q}$	{A1} (4)
		Total 14 marks

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Part	Mark	Notes
(a)	M1	For finding the vector $\overrightarrow{OB}$: $\left(\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}\right) = 5\mathbf{p} + 3\mathbf{q}$ **This must be correct.**
	M1	For the correct vector statement for $\overrightarrow{DC}$: $\overrightarrow{DC} = \left(\overrightarrow{DO} + \overrightarrow{OC}\right) = \overrightarrow{DO} + \frac{3}{2}\overrightarrow{OB}$ This mark can be implied by a correct (unsimplified) vector.
	A1	For the correct simplified $\overrightarrow{DC}$ in terms of a single **p** and a single **q** only $\overrightarrow{DC} = -3\mathbf{p} + \frac{3}{2}(5\mathbf{p} + 3\mathbf{q}) = \frac{9}{2}(\mathbf{p} + \mathbf{q})$ accept $\overrightarrow{DC} = \frac{9\mathbf{p}}{2} + \frac{9\mathbf{q}}{2}$ oe
(b)	M1	For stating $\overrightarrow{OF} = \overrightarrow{OD} + \lambda\overrightarrow{DC}$ (or any other variable in place of λ)
	A1	For the vector $\overrightarrow{OF} = 3\mathbf{p} + \frac{9}{2}\lambda(\mathbf{p} + \mathbf{q})$
	M1	For stating $\overrightarrow{OF} = \overrightarrow{OA} + \mu\overrightarrow{AB}$ (or any other variable in place of μ)
	A1	For the vector $\overrightarrow{OF} = 5\mathbf{p} + 3\mu\mathbf{q}$
	M1	Collects like terms correctly, equates coefficients correctly and attempts to solve their two equations in λ and μ They must achieve a value for μ or λ for this mark $3\mathbf{p} + \frac{9}{2}\lambda(\mathbf{p} + \mathbf{q}) = 5\mathbf{p} + 3\mu\mathbf{q} \Rightarrow \mathbf{p}\left(3 + \frac{9}{2}\lambda\right) + \frac{9}{2}\lambda\mathbf{q} = 5\mathbf{p} + 3\mu\mathbf{q}$ $3 + \frac{9}{2}\lambda = 5$ and $\frac{9}{2}\lambda = 3\mu$ $\Rightarrow \mu = \frac{2}{3}$ or $\lambda = \frac{4}{9}$
	A1	For the correct $\mu = \frac{2}{3}$ or $\lambda = \frac{4}{9}$
	A1	For the correct vector $\overrightarrow{OF} = 5\mathbf{p} + 2\mathbf{q}$
		Part (b) can be solved in different ways. These are the general principles of marking this part question. **M1** – for any correct vector statement leading to $\overrightarrow{OF}$ which introduces a parameter. **A1** – for one correct vector $\overrightarrow{OF}$ **M1** - For any correct second vector statement for $\overrightarrow{OF}$ which can be used with the first vector and introduces a second parameter. **A1** – For a second correct vector $\overrightarrow{OF}$ **M1A1A1** – as main scheme above
(c)	M1	For one vector for $\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$

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	M1	For a second vector for $\overrightarrow{OG}=\left(\overrightarrow{OF}+\overrightarrow{FG}\right)=3\mathbf{p}+2(\mathbf{p}+\mathbf{q})+5\beta\mathbf{p}=\left(5\mathbf{p}+5\beta\mathbf{p}+2\mathbf{q}\right)$ ft their $\overrightarrow{OF}$ Accept simplified or unsimplified.
	M1	Collects like terms correctly, equates coefficients correctly to find the value of α $\alpha(5\mathbf{p}+3\mathbf{q})=5\mathbf{p}+5\beta\mathbf{p}+2\mathbf{q}\Rightarrow 5\mathbf{p}\alpha+3\mathbf{q}\alpha=(5+5\beta)\mathbf{p}+2\mathbf{q}$ $\Rightarrow 3\alpha=2\Rightarrow\alpha=\frac{2}{3}$
	A1	For the correct vector $\overrightarrow{OG}=\frac{10}{3}\mathbf{p}+2\mathbf{q}$
	ALT	
	M1	For the vector for $\overrightarrow{OG}=\alpha(5\mathbf{p}+3\mathbf{q})$
	M1	For the vector $\overrightarrow{FG}=\left(\overrightarrow{FO}+\overrightarrow{OG}\right)=-5\mathbf{p}-2\mathbf{q}+\alpha\ 5\mathbf{p}+3\mathbf{q}$ **Note:** ft their $-\left(\overrightarrow{OF}\right)$ from part (b)
	M1	As FG is parallel to OA there is no $\mathbf{q}$ component in $\overrightarrow{FG}\Rightarrow 3\alpha=2\Rightarrow\alpha=\frac{2}{3}$
	A1	For the correct vector $\overrightarrow{OG}=\frac{10}{3}\mathbf{p}+2\mathbf{q}$

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10 (a) (i)	$\mathbf{a}+\mathbf{c}$	B1
(ii)	$\frac{1}{2}(\mathbf{c}-\mathbf{a})$	B1 (2)
(b)	$\overrightarrow{OX} = OA + AM + \lambda MN$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda(\frac{1}{2}\mathbf{c}-\frac{1}{2}\mathbf{a})$	M1 A1
	$\mu(\mathbf{a}+\mathbf{c})$	B1
	$\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda\left(\frac{1}{2}\mathbf{c}-\frac{1}{2}\mathbf{a}\right)=\mu(\mathbf{a}+\mathbf{c})$	M1
	$1-\frac{1}{2}\lambda=\mu \quad \frac{1}{2}+\frac{1}{2}\lambda=\mu$	M1
	$1-\frac{1}{2}\lambda=\frac{1}{2}+\frac{1}{2}\lambda$	M1
	$\lambda=\frac{1}{2} \quad \mu=\frac{3}{4}$	A1 A1 (8)
(c)	Triangle $XBN=\frac{1}{8}$ of $\frac{1}{2}$ the parallelogram Quadrilateral $OXNC=\frac{7}{8}$ of $\frac{1}{2}$ the parallelogram So Quadrilateral $OXNC=\frac{7}{16}$ of the parallelogram $\therefore$ 7 : 16	M1 M1 A1 (3) **[13]**

Part	Mark	Additional Guidance
(a)(i)	B1	For the correct vector $\mathbf{a}+\mathbf{c}$
(ii)	B1	For the correct vector $\frac{1}{2}(\mathbf{c}-\mathbf{a})$
(b)	M1	For the correct vector statement $\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AM} + \lambda\overrightarrow{MN}$
	A1	For the correct vector (need not be simplified) $\overrightarrow{OX}=\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$ or $\overrightarrow{OX}=\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda\left(\frac{\mathbf{c}}{2}-\frac{\mathbf{a}}{2}\right)$
	B1ft	For $\overrightarrow{OX}=\mu(\mathbf{a}+\mathbf{c})$ ft their $\overrightarrow{OB}=$'$\mathbf{a}+\mathbf{c}$'
	M1	For equating their two vector statements for $\overrightarrow{OX}$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})=\mu(\mathbf{a}+\mathbf{c})$
	M1	For equating coefficients of $\mathbf{a}$ and $\mathbf{c}$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})=\mu(\mathbf{a}-\mathbf{c})\Rightarrow\mathbf{a}\left(1-\frac{\lambda}{2}\right)+\mathbf{c}\left(\frac{1}{2}+\frac{\lambda}{2}\right)=\mu\mathbf{a}+\mu\mathbf{c}$ $\Rightarrow\mu=1-\frac{\lambda}{2},\ \mu=\frac{1}{2}+\frac{\lambda}{2}$
	M1	For attempting to solve their two simultaneous equations in terms of λ and μ. $1-\frac{\lambda}{2}=\frac{1}{2}+\frac{\lambda}{2}\Rightarrow\lambda=\ldots\Rightarrow\mu=\ldots$ or $1-\mu=\mu-\frac{1}{2}\Rightarrow\mu=\ldots\Rightarrow\lambda=\ldots$

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	A1	For **either** $\lambda = \frac{1}{2}$ **or** $\mu = \frac{3}{4}$
	A1	For **both** $\lambda = \frac{1}{2}$ **and** $\mu = \frac{3}{4}$
	ALT	
	M1	For the correct vector statement $\overrightarrow{MX} = \overrightarrow{MO} + \overrightarrow{OX}$
	A1	For the correct vector (need not be simplified) $\overrightarrow{MX} = -\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c})$
	B1ft	$\overrightarrow{MX} = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$ ft their $\overrightarrow{MN} = '\frac{1}{2}(\mathbf{c}-\mathbf{a})'$
	M1	For equating the two vector statements for $\overrightarrow{MX}$ $-\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c}) = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$
	M1	For equating coefficients of $\mathbf{a}$ and $\mathbf{c}$ $-\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c}) = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a}) \Rightarrow \mathbf{c}\left(-\frac{1}{2}+\mu\right) + \mathbf{a}(\mu-1) = \mathbf{c}\frac{\lambda}{2} - \mathbf{a}\frac{\lambda}{2}$ $\Rightarrow \frac{\lambda}{2} = \mu - \frac{1}{2}$ and $-\frac{\lambda}{2} = \mu - 1$
	M1	For attempting to solve their two simultaneous equations in terms of λ and μ. $\mu - \frac{1}{2} = -(\mu - 1) \Rightarrow \mu = \left(\frac{3}{4}\right)$ $\frac{\lambda}{2} = 1 - '\frac{3}{4}' \Rightarrow \lambda = \left(\frac{1}{2}\right)$
	A1	For **either** $\lambda = \frac{1}{2}$ **or** $\mu = \frac{3}{4}$
	A1	For **both** $\lambda = \frac{1}{2}$ **and** $\mu = \frac{3}{4}$
(c)	M1	For area of $\Delta XBN = \frac{1}{8}\Delta OBC$ so $\frac{1}{8}$ of $\frac{1}{2}$ of the area of parallelogram $OABC$ $\left(\Delta OBC = \frac{1}{2} \times OB \times BC \times \sin \angle XBN\right.$ $\left.\Delta XBN = \frac{1}{2} \times \frac{1}{4} OB \times \frac{1}{2} BC \times \sin \angle XBN = \frac{1}{8}\Delta OBC\right)$
	M1	Therefore Quadrilateral $OXNC = \frac{7}{8}$ of $\frac{1}{2}$ of the area of parallelogram $OABC$ ft their fraction from the first M mark provided it is $< \frac{1}{2}$
	A1	Quadrilateral $OXNC = \frac{7}{16}$ of the area of parallelogram $OABC$ so ratio is 7:16

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Question number	Scheme	Marks

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Answer 6 of 18 · Question 8 part a · 1 mark

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8 (a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \mathbf{a} + \mathbf{b}$	B1 [1]
(b)	$\overrightarrow{OC} = 2\mathbf{b}$ $\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC} = -(\mathbf{a} + \mathbf{b}) + 2\mathbf{b} = \mathbf{b} - \mathbf{a}$	B1 M1A1 [3]
(c)	$\overrightarrow{OM} = \overrightarrow{OB} + \overrightarrow{BM} = \mathbf{a} + \mathbf{b} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}$ or $\frac{1}{3}(\mathbf{a} + 5\mathbf{b})$	 M1A1ft [2]
(d)	$\overrightarrow{OY} = \mu\left(\frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}\right) = \frac{\mu\mathbf{a}}{3} + \frac{5\mu\mathbf{b}}{3}$ $\overrightarrow{OY} = \overrightarrow{OA} + \overrightarrow{AY} = \boldsymbol{a} + \lambda\boldsymbol{b}$ $\Rightarrow \frac{\mu}{3} = 1$ and $\frac{5\mu}{3} = \lambda$ Solves simultaneous equations by any method $\mu = 3,\ \lambda = 5$ $AB : BY = 1 : 4$ **ALT** $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ $\Rightarrow -1 + \frac{1}{3}\mu = 0$ and $\lambda = -1 + \frac{5}{3}\mu$ $\mu = 3, \lambda = 4$ $AB : BY = 1 : 4$	M1 M1 M1 M1 A1 [5] [M1 M1 M1 M1 A1]
(e)	$10 = \frac{1}{2}ab\sin 60^\circ \Rightarrow ab = \frac{40}{\sqrt{3}} \Rightarrow a = \frac{40}{b\sqrt{3}}$ $\text{Area} = \frac{1}{2}a \times 5b\sin 120^\circ = \frac{1}{2} \times \frac{40}{b\sqrt{3}} \times 5b\sin 120^\circ$ $\text{Area} = 50$ **ALT** $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2} \times h \times 5}{\frac{1}{2} \times h \times 1} = \frac{5}{1}$ $\text{Area } OAY = 5 \times \text{Area } OAB$ $\text{Area } OAB = 10$ $\text{Area} = 50$	M1A1 dM1 A1 [4] [M1 A1 dM1 A1]
		Total 15 marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 6 of 18 · Question 8 part a · 1 mark

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Part	Mark	Notes
(a)	B1	For a correct expression for $\overrightarrow{OB}$ in terms of **a** and **b**
(b)	B1	For a correct expression for $\overrightarrow{OC}$
	M1	For a correct vector statement for $\overrightarrow{BC}$: $\overrightarrow{BC}=\overrightarrow{BO}+\overrightarrow{OC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$. Vector statement must be suitable for substitution to find $\overrightarrow{BC}$
	A1	For the correct simplified $\overrightarrow{BC}$ in terms of a single **a** and **b** only. $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ If answer $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ seen without wrong working then award B1M1A1.
(c)	M1	For a correct vector statement for $\overrightarrow{OM}$: $\overrightarrow{OM}=\overrightarrow{OB}+\frac{2}{3}\overrightarrow{BC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$ and their $\overrightarrow{BC}$.
	A1ft	For the correct simplified using their $\overrightarrow{OM}$ in terms of a single **a** and **b** only. $\overrightarrow{OM} = \frac{\boldsymbol{a}}{3} + \frac{5\boldsymbol{b}}{3}$ or $\frac{1}{3}(\boldsymbol{a} + 5\boldsymbol{b})$
(d)	M1 M1	M1 for one correct statement of route for $\overrightarrow{OY}$ M1 for second correct statement of route for $\overrightarrow{OY}$ $\overrightarrow{OY} = \mu\overrightarrow{OM}$ (or any other variable in place of μ) $\overrightarrow{OY} = \left(\overrightarrow{OA} + \overrightarrow{AY}\right) = \mathbf{a} + \lambda\mathbf{b}$ (or any other variable in place of λ, provided this is different to their μ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$
ALT – use of $\overrightarrow{BY}$ route, use also for $\overrightarrow{AY}$ route		
	M1 M1	M1 for one correct statement of route for $\overrightarrow{BY}$ M1 for second correct statement of route for $\overrightarrow{BY}$ $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ (or any other variable in place of λ) $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ (or any other variable in place of μ provided this is different to their λ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 6 of 18 · Question 8 part a · 1 mark

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(e)	**M1**	For use of the **correct** formula for area of a triangle with $60°$ and correct value of $\sin 60°$ $10 = \frac{1}{2}ab\sin 60° = ab \times \frac{\sqrt{3}}{4}$
	A1	For correct expression for ab or a $ab = \frac{40}{\sqrt{3}}$ or $a = \frac{40}{b\sqrt{3}}$
	dM1	For use of the **correct** formula for area of a triangle with 120° and attempt to substitute for ab or a. $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times '\frac{40}{b\sqrt{3}}' \times '5'b\sin 120°$ or $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times ab \times '5'\sin 120° = \frac{1}{2} \times '\frac{40}{\sqrt{3}}' \times '5'\sin 120°$ Dependent on the first M awarded. Allow use of their λ from part (d).
	A1	For the correct area $\text{Area} = 50$
ALT – use of ratios of areas		
	M1	For use of their ratio $AB:BY$ to write an equation linking area OAY and area OAB $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2}\times h\times(1+'4')}{\frac{1}{2}\times h\times 1} = \frac{'5'}{1}$
	A1	For correct relationship between area OAY and area OAB
	dM1	For a correct method to find the area of OAB Dependent on first M mark being awarded.
	A1	For the correct area $\text{Area} = 50$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 6 of 18 · Question 8 part a · 1 mark

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Question number	Scheme	Marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 7 of 18 · Question 8 part b · 3 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

8 (a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \mathbf{a} + \mathbf{b}$	B1 [1]
(b)	$\overrightarrow{OC} = 2\mathbf{b}$ $\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC} = -(\mathbf{a} + \mathbf{b}) + 2\mathbf{b} = \mathbf{b} - \mathbf{a}$	B1 M1A1 [3]
(c)	$\overrightarrow{OM} = \overrightarrow{OB} + \overrightarrow{BM} = \mathbf{a} + \mathbf{b} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}$ or $\frac{1}{3}(\mathbf{a} + 5\mathbf{b})$	 M1A1ft [2]
(d)	$\overrightarrow{OY} = \mu\left(\frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}\right) = \frac{\mu\mathbf{a}}{3} + \frac{5\mu\mathbf{b}}{3}$ $\overrightarrow{OY} = \overrightarrow{OA} + \overrightarrow{AY} = \boldsymbol{a} + \lambda\boldsymbol{b}$ $\Rightarrow \frac{\mu}{3} = 1$ and $\frac{5\mu}{3} = \lambda$ Solves simultaneous equations by any method $\mu = 3,\ \lambda = 5$ $AB : BY = 1 : 4$ **ALT** $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ $\Rightarrow -1 + \frac{1}{3}\mu = 0$ and $\lambda = -1 + \frac{5}{3}\mu$ $\mu = 3, \lambda = 4$ $AB : BY = 1 : 4$	M1 M1 M1 M1 A1 [5] [M1 M1 M1 M1 A1]
(e)	$10 = \frac{1}{2}ab\sin 60^\circ \Rightarrow ab = \frac{40}{\sqrt{3}} \Rightarrow a = \frac{40}{b\sqrt{3}}$ $\text{Area} = \frac{1}{2}a \times 5b\sin 120^\circ = \frac{1}{2} \times \frac{40}{b\sqrt{3}} \times 5b\sin 120^\circ$ $\text{Area} = 50$ **ALT** $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2} \times h \times 5}{\frac{1}{2} \times h \times 1} = \frac{5}{1}$ $\text{Area } OAY = 5 \times \text{Area } OAB$ $\text{Area } OAB = 10$ $\text{Area} = 50$	M1A1 dM1 A1 [4] [M1 A1 dM1 A1]
		Total 15 marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 7 of 18 · Question 8 part b · 3 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For a correct expression for $\overrightarrow{OB}$ in terms of **a** and **b**
(b)	B1	For a correct expression for $\overrightarrow{OC}$
	M1	For a correct vector statement for $\overrightarrow{BC}$: $\overrightarrow{BC}=\overrightarrow{BO}+\overrightarrow{OC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$. Vector statement must be suitable for substitution to find $\overrightarrow{BC}$
	A1	For the correct simplified $\overrightarrow{BC}$ in terms of a single **a** and **b** only. $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ If answer $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ seen without wrong working then award B1M1A1.
(c)	M1	For a correct vector statement for $\overrightarrow{OM}$: $\overrightarrow{OM}=\overrightarrow{OB}+\frac{2}{3}\overrightarrow{BC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$ and their $\overrightarrow{BC}$.
	A1ft	For the correct simplified using their $\overrightarrow{OM}$ in terms of a single **a** and **b** only. $\overrightarrow{OM} = \frac{\boldsymbol{a}}{3} + \frac{5\boldsymbol{b}}{3}$ or $\frac{1}{3}(\boldsymbol{a} + 5\boldsymbol{b})$
(d)	M1 M1	M1 for one correct statement of route for $\overrightarrow{OY}$ M1 for second correct statement of route for $\overrightarrow{OY}$ $\overrightarrow{OY} = \mu\overrightarrow{OM}$ (or any other variable in place of μ) $\overrightarrow{OY} = \left(\overrightarrow{OA} + \overrightarrow{AY}\right) = \mathbf{a} + \lambda\mathbf{b}$ (or any other variable in place of λ, provided this is different to their μ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$
ALT – use of $\overrightarrow{BY}$ route, use also for $\overrightarrow{AY}$ route		
	M1 M1	M1 for one correct statement of route for $\overrightarrow{BY}$ M1 for second correct statement of route for $\overrightarrow{BY}$ $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ (or any other variable in place of λ) $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ (or any other variable in place of μ provided this is different to their λ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 7 of 18 · Question 8 part b · 3 marks

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(e)	**M1**	For use of the **correct** formula for area of a triangle with $60°$ and correct value of $\sin 60°$ $10 = \frac{1}{2}ab\sin 60° = ab \times \frac{\sqrt{3}}{4}$
	A1	For correct expression for ab or a $ab = \frac{40}{\sqrt{3}}$ or $a = \frac{40}{b\sqrt{3}}$
	dM1	For use of the **correct** formula for area of a triangle with $120°$ and attempt to substitute for ab or a. $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times '\frac{40}{b\sqrt{3}}' \times '5'b\sin 120°$ or $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times ab \times '5'\sin 120° = \frac{1}{2} \times '\frac{40}{\sqrt{3}}' \times '5'\sin 120°$ Dependent on the first M awarded. Allow use of their λ from part (d).
	A1	For the correct area Area $= 50$
ALT – use of ratios of areas		
	M1	For use of their ratio $AB:BY$ to write an equation linking area OAY and area OAB $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2}\times h\times(1+'4')}{\frac{1}{2}\times h\times 1} = \frac{'5'}{1}$
	A1	For correct relationship between area OAY and area OAB
	dM1	For a correct method to find the area of OAB Dependent on first M mark being awarded.
	A1	For the correct area Area $= 50$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 7 of 18 · Question 8 part b · 3 marks

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Question number	Scheme	Marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 8 of 18 · Question 11 part a.i · 5 marks

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11(a)	(i) $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$	B1	
	(ii) $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC} = \mathbf{a} + \frac{1}{2}\left(-\mathbf{a} + \frac{1}{4}\mathbf{b}\right) = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$	M1A1	
	(iii) $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD} = -\mathbf{b} + \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$	M1A1	(5)
(b)	$\overrightarrow{BE} = k\overrightarrow{BD} = k\left(\frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}\right)$	M1	
	$\overrightarrow{BE} = -\mathbf{b} + \overrightarrow{OE} = -\mathbf{b} + \lambda\mathbf{a}$	M1	
	$\frac{7}{8}k = 1 \Rightarrow k = \frac{8}{7}$	M1	
	$\frac{k}{2} = \lambda \Rightarrow \lambda = \frac{4}{7}$	A1	(4)
(c)	$\triangle OAC = \frac{1}{4}\triangle OAB$	M1	
	$\triangle OEB = "\frac{4}{7}"\triangle OAB$	A1 ft	
	$\frac{\triangle OAC}{\triangle OEB} = \frac{1}{4} \times "\frac{7}{4}" = \frac{7}{16}$	M1	
	$\mu = \frac{7}{16}$	A1	(4)
			[13]

Part	Mark	Notes
(a)(i)	B1	For the correct vector $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$
(ii)	M1	For the correct vector statement $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC}$ or $\overrightarrow{OD} = \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CA}$
	A1	For the correct **simplified** vector $\overrightarrow{OD} = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$
(iii)	M1	For the correct vector statement $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD}$ or $\overrightarrow{BD} = \overrightarrow{BC} + \overrightarrow{CD}$
	A1	For the correct **simplified** vector $\overrightarrow{BD} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 8 of 18 · Question 11 part a.i · 5 marks

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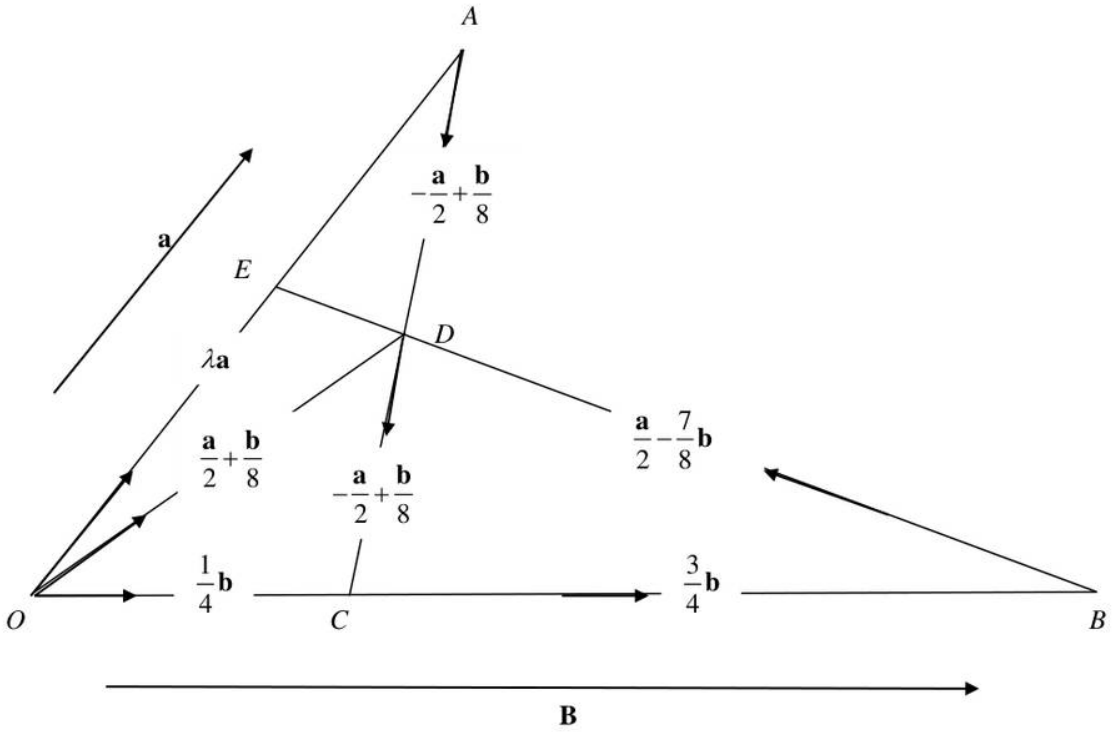
In parts (b) and (c) you must follow through the vectors they have found in part (a)		
(b)	M1	States or implies two paths that will lead to a solution. For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}$ and $\overrightarrow{BE}=\overrightarrow{BO}+\overrightarrow{OE}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\overrightarrow{OD}+\overrightarrow{DE}$ **This is a B mark in Epen**
	M1	For writing their paths as vectors in terms of **a**, **b** λ **and another constant** (e.g. k or μ) For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}=k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ and $\overrightarrow{BE}=-\mathbf{b}+\lambda\mathbf{a}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ OR $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$
	dM1	Equates coefficients of their **two** expressions and attempts to find the value of λ: In triangle OEB • $k\frac{1}{2}\mathbf{a}-k\frac{7}{8}\mathbf{b}=\lambda\mathbf{a}-\mathbf{b}$ $\Rightarrow k\frac{7}{8}=1\Rightarrow k=\frac{8}{7}$ and $\frac{k}{2}=\lambda\Rightarrow\lambda=...$ or in triangle OED, there are two possible expressions for $\overrightarrow{DE}$ • $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ $\Rightarrow k=\frac{1}{8}$ and $\lambda=\frac{1}{2}+\frac{1}{8}\lambda\Rightarrow\lambda=...$ Or $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ $\Rightarrow\frac{7}{8}k=\frac{1}{8}\Rightarrow k=\frac{1}{7}$ and $\lambda=\frac{1}{2}+\frac{1}{7}k\Rightarrow\lambda...$
	A1	For the correct value of λ $\lambda=\frac{4}{7}$
(c)	M1	For stating that $\triangle OAC=\frac{1}{4}\triangle OAB$ or $4\triangle OAC=\triangle OAB$ **1**
	M1	For stating $\triangle OEB="\lambda"\triangle OAB\Rightarrow\triangle OEB="\frac{4}{7}"\triangle OAB$ **2** **This is an A mark in Epen**
	M1	For dividing **1** by **2** $=\frac{\triangle OAC}{\triangle OEB}=\frac{1}{4}\times"\frac{7}{4}"=...$
	A1	For $\mu=\frac{7}{16}$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 8 of 18 · Question 11 part a.i · 5 marks

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USEFUL SKETCH



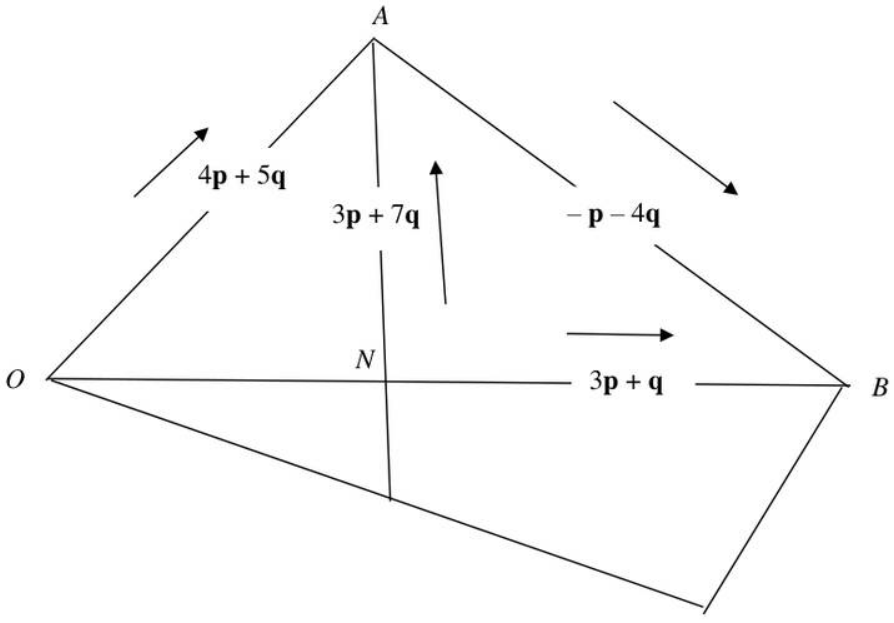
Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 9 of 18 · Question 11 part a · 3 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

11(a)	$\overrightarrow{OM} = \frac{1}{2}\overrightarrow{OC} = \mathbf{p} - 2\mathbf{q}$ $\overrightarrow{MA} = -\overrightarrow{OM} + \overrightarrow{OA}$ $\overrightarrow{MA} = -(\mathbf{p} - 2\mathbf{q}) + (4\mathbf{p} + 5\mathbf{q}) = 3\mathbf{p} + 7\mathbf{q}$	B1 M1 A1 [3]
(b)	$\overrightarrow{MN} = \lambda(3\mathbf{p} + 7\mathbf{q})$ $\overrightarrow{MN} = -(\mathbf{p} - 2\mathbf{q}) + \mu(3\mathbf{p} + \mathbf{q})$ $\lambda(3\mathbf{p} + 7\mathbf{q}) = -(\mathbf{p} - 2\mathbf{q}) + \mu(3\mathbf{p} + \mathbf{q})$ $\Rightarrow 3\lambda\mathbf{p} + 7\lambda\mathbf{q} = (-1 + 3\mu)\mathbf{p} + (2 + \mu)\mathbf{q}$ $3\lambda = -1 + 3\mu$ $7\lambda = 2 + \mu$ $\Rightarrow 18\lambda = 7 \Rightarrow \lambda = \frac{7}{18}$ $MN : NA = 7 : 11$	M1 M1 M1 dM1 A1 A1 [6]
		Total 9 marks

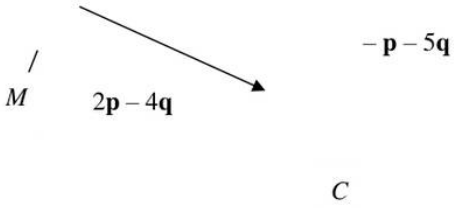
Useful sketch



Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 9 of 18 · Question 11 part a · 3 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract



Part	Mark	Notes
(a)	B1	For finding either vector $\overrightarrow{OM} = \mathbf{p} - 2\mathbf{q}$ or $\overrightarrow{MO} = -\mathbf{p} + 2\mathbf{q}$ This may be embedded in their working for $\overrightarrow{MA}$
	M1	For the correct vector statement $\overrightarrow{MA} = -\overrightarrow{OM} + \overrightarrow{OA}$ or $\overrightarrow{MA} = -\frac{1}{2}\overrightarrow{OC} + \overrightarrow{OA}$
	A1	For the correct simplified vector
(b)	**General principle of marking part (b)** • First two M marks are for two vector statements for $\overrightarrow{MN}$ or $\overrightarrow{AN}$ that will allow them to be equated. Note that they can use any path that involves either of these two vectors. • The third M mark is for equating coefficients and forming a pair of simultaneous equations. • The final M mark is for solving their simultaneous equations.	
	M1	For one vector for $\overrightarrow{MN}$ or $\overrightarrow{AN}$ involving a constant One example is: $\overrightarrow{AN} = K\overrightarrow{AM} = K(-3\mathbf{p} - 7\mathbf{q})$
	M1	For a second vector for $\overrightarrow{MN}$ or $\overrightarrow{AN}$ following a different path involving a different constant. One example is: $\overrightarrow{AN} = \overrightarrow{AO} + L\overrightarrow{OB} = -4\mathbf{p} - 5\mathbf{q} + L(3\mathbf{p} + \mathbf{q})$
	M1	For equating the two vectors and forming a pair of simultaneous linear equations, both of which must be in terms of λ and μ
	dM1	For an attempt to solve their linear equations. Allow up to one processing error. This mark is dependent on the previous M mark.
	A1	For the value of $\lambda = \frac{7}{18}$ μ is not required but the value is $\frac{13}{18}$
	A1	For the correct ratio $MN : NA = 7 : 11$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 9 of 18 · Question 11 part a · 3 marks

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Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 10 of 18 · Question 9 part b · 2 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

9(a)	$\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB} \Rightarrow \overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$	M1A1 [2]
(b)	$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} \Rightarrow \overrightarrow{OP} = 2\mathbf{a} + \frac{1}{2}(4\mathbf{b} - 2\mathbf{a}) = \mathbf{a} + 2\mathbf{b}$	M1A1 [2]
(c)	$\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ}$ $\overrightarrow{OQ} = \frac{3}{4}(\overrightarrow{OP})$ $\overrightarrow{AQ} = -2\mathbf{a} + \frac{3}{4}(\mathbf{a} + 2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$	M1 B1 A1 [3]
(d)	$\overrightarrow{OR} = \lambda 4\mathbf{b}$ or $\overrightarrow{OR} = \phi\mathbf{b}$ $\overrightarrow{OR} = \overrightarrow{OA} + \mu\overrightarrow{AQ} = 2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right)$ $2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right) = \lambda 4\mathbf{b}$ $\mathbf{a}\left(2 - \frac{5\mu}{4}\right) + \frac{3\mu}{2}\mathbf{b} = \lambda 4\mathbf{b} \Rightarrow \frac{3\mu}{2} = 4\lambda$ and $2 - \frac{5\mu}{4} = 0$ $2 - \frac{5\mu}{4} = 0 \Rightarrow \mu = \left(\frac{8}{5}\right)$ $\frac{3}{2} \times \frac{8}{5} = 4\lambda \Rightarrow \lambda = \frac{12}{20}, \left(\frac{3}{5}\right)$ $OR : RB = 3 : 2$	M1 M1 M1 M1A1 A1 [6]
	Total	**13 marks**

Part	Mark	Scheme
(a)	M1	For the correct vector statement $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB}$
	A1	$\overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$
(b)	M1	For a correct vector statement e.g $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$
	A1	For the correct simplified vector $\overrightarrow{OP} = 2\mathbf{a} + (2\mathbf{b} - \mathbf{a}) = \mathbf{a} + 2\mathbf{b}$
(c)	M1	For a correct vector statement e.g., $\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ}$ or $\overrightarrow{AQ} = \overrightarrow{AP} + \overrightarrow{PQ}$
	B1	For the correct vector statement for $\overrightarrow{OQ}$ $\overrightarrow{OQ} = \frac{3}{4}\left(\overrightarrow{OP}\right)$ or $\overrightarrow{PQ} = -\frac{1}{4}\left(\overrightarrow{OP}\right)$
	A1	For the correct simplified vector

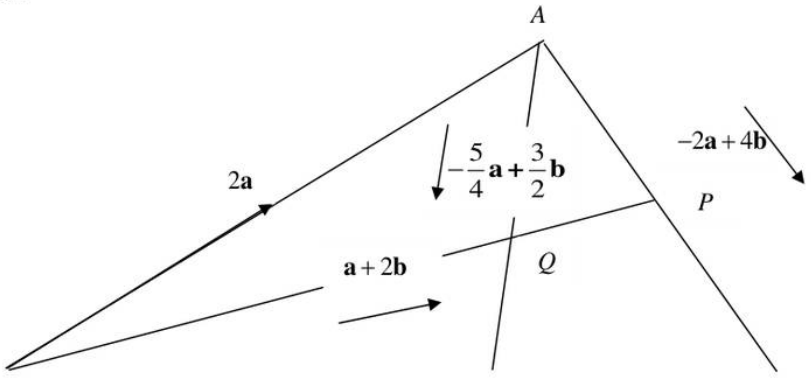
Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 10 of 18 · Question 9 part b · 2 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

		$\overrightarrow{AQ} = -2\mathbf{a} + \frac{3}{4}(\mathbf{a}+2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$ or $\overrightarrow{AQ} = \frac{1}{2}(-2\mathbf{a}+4\mathbf{b}) - \frac{1}{4}(\mathbf{a}+2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$
(d)	In this part there is more than one route to finding the required ratio, but any route must involve $\overrightarrow{OR}$ or $\overrightarrow{RB}$ As the general principle: The first M mark is for one vector using a parameter. The second M mark is for a second vector using a different parameter. The third M mark is for equating coefficients and forming two equations in both parameters. The fourth M mark is for solving their equations. Check their working and do not allow an erroneous method. The following is using one path. Please trace their path on the sketch to check that it is valid. The path they use must involve the vector $\overrightarrow{OR}$ or $\overrightarrow{RB}$ eg., $\overrightarrow{AR} = \overrightarrow{AO} + \overrightarrow{OR}$ or $\overrightarrow{AR} = \overrightarrow{AB} + \overrightarrow{BR}$	
	M1	For the correct statement $\overrightarrow{OR} = \lambda 4\mathbf{b}$ or $\overrightarrow{OR} = \phi\mathbf{b}$
	M1	For the correct statement for $\overrightarrow{OR}$ $\overrightarrow{OR} = \overrightarrow{OA} + \mu\overrightarrow{AQ} = 2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right)$
	M1	For equating the two statements for $\overrightarrow{OR}$ and equating coefficients $2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right) = \lambda 4\mathbf{b}$ $\mathbf{a}\left(2 - \frac{5\mu}{4}\right) + \frac{3\mu}{2}\mathbf{b} = \lambda 4\mathbf{b} \Rightarrow \frac{3\mu}{2} = 4\lambda$ and $2 - \frac{5\mu}{4} = 0$
	M1	For solving their two simultaneous equations **correctly** to find the value of λ (the parameter for OR) $2 - \frac{5\mu}{4} = 0 \Rightarrow \mu = \frac{8}{5}$ $\frac{3}{2} \times \frac{8}{5} = 4\lambda \Rightarrow \lambda = \ldots$
	A1	For the correct value of λ $\lambda = \frac{12}{20} = \left(\frac{3}{5}\right)$
	A1	For the correct ratio $OR : RB = 3 : 2$

Useful Sketch



Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 10 of 18 · Question 9 part b · 2 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract



Question	Scheme	Mark

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 11 of 18 · Question 9 part c · 3 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

9(a)	$\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB} \Rightarrow \overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$	M1A1 [2]
(b)	$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} \Rightarrow \overrightarrow{OP} = 2\mathbf{a} + \frac{1}{2}(4\mathbf{b} - 2\mathbf{a}) = \mathbf{a} + 2\mathbf{b}$	M1A1 [2]
(c)	$\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ}$ $\overrightarrow{OQ} = \frac{3}{4}(\overrightarrow{OP})$ $\overrightarrow{AQ} = -2\mathbf{a} + \frac{3}{4}(\mathbf{a} + 2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$	M1 B1 A1 [3]
(d)	$\overrightarrow{OR} = \lambda 4\mathbf{b}$ or $\overrightarrow{OR} = \phi\mathbf{b}$ $\overrightarrow{OR} = \overrightarrow{OA} + \mu\overrightarrow{AQ} = 2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right)$ $2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right) = \lambda 4\mathbf{b}$ $\mathbf{a}\left(2 - \frac{5\mu}{4}\right) + \frac{3\mu}{2}\mathbf{b} = \lambda 4\mathbf{b} \Rightarrow \frac{3\mu}{2} = 4\lambda$ and $2 - \frac{5\mu}{4} = 0$ $2 - \frac{5\mu}{4} = 0 \Rightarrow \mu = \left(\frac{8}{5}\right)$ $\frac{3}{2} \times \frac{8}{5} = 4\lambda \Rightarrow \lambda = \frac{12}{20}, \left(\frac{3}{5}\right)$ $OR : RB = 3 : 2$	M1 M1 M1 M1A1 A1 [6]
	Total	**13 marks**

Part	Mark	Scheme
(a)	M1	For the correct vector statement $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB}$
	A1	$\overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$
(b)	M1	For a correct vector statement e.g $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$
	A1	For the correct simplified vector $\overrightarrow{OP} = 2\mathbf{a} + (2\mathbf{b} - \mathbf{a}) = \mathbf{a} + 2\mathbf{b}$
(c)	M1	For a correct vector statement e.g., $\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ}$ or $\overrightarrow{AQ} = \overrightarrow{AP} + \overrightarrow{PQ}$
	B1	For the correct vector statement for $\overrightarrow{OQ}$ $\overrightarrow{OQ} = \frac{3}{4}\left(\overrightarrow{OP}\right)$ or $\overrightarrow{PQ} = -\frac{1}{4}\left(\overrightarrow{OP}\right)$
	A1	For the correct simplified vector

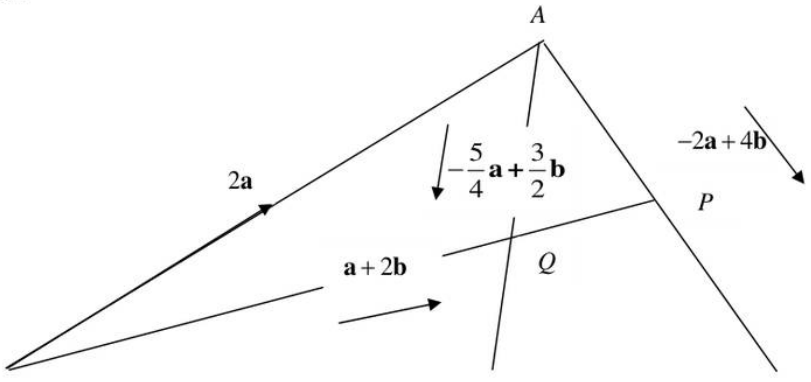
Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 11 of 18 · Question 9 part c · 3 marks

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		$\overrightarrow{AQ} = -2\mathbf{a} + \frac{3}{4}(\mathbf{a}+2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$ or $\overrightarrow{AQ} = \frac{1}{2}(-2\mathbf{a}+4\mathbf{b}) - \frac{1}{4}(\mathbf{a}+2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$
(d)	In this part there is more than one route to finding the required ratio, but any route must involve $\overrightarrow{OR}$ or $\overrightarrow{RB}$ As the general principle: The first M mark is for one vector using a parameter. The second M mark is for a second vector using a different parameter. The third M mark is for equating coefficients and forming two equations in both parameters. The fourth M mark is for solving their equations. Check their working and do not allow an erroneous method. The following is using one path. Please trace their path on the sketch to check that it is valid. The path they use must involve the vector $\overrightarrow{OR}$ or $\overrightarrow{RB}$ eg., $\overrightarrow{AR} = \overrightarrow{AO} + \overrightarrow{OR}$ or $\overrightarrow{AR} = \overrightarrow{AB} + \overrightarrow{BR}$	
	M1	For the correct statement $\overrightarrow{OR} = \lambda 4\mathbf{b}$ or $\overrightarrow{OR} = \phi\mathbf{b}$
	M1	For the correct statement for $\overrightarrow{OR}$ $\overrightarrow{OR} = \overrightarrow{OA} + \mu\overrightarrow{AQ} = 2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right)$
	M1	For equating the two statements for $\overrightarrow{OR}$ and equating coefficients $2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right) = \lambda 4\mathbf{b}$ $\mathbf{a}\left(2 - \frac{5\mu}{4}\right) + \frac{3\mu}{2}\mathbf{b} = \lambda 4\mathbf{b} \Rightarrow \frac{3\mu}{2} = 4\lambda$ and $2 - \frac{5\mu}{4} = 0$
	M1	For solving their two simultaneous equations **correctly** to find the value of λ (the parameter for OR) $2 - \frac{5\mu}{4} = 0 \Rightarrow \mu = \frac{8}{5}$ $\frac{3}{2} \times \frac{8}{5} = 4\lambda \Rightarrow \lambda = \ldots$
	A1	For the correct value of λ $\lambda = \frac{12}{20} = \left(\frac{3}{5}\right)$
	A1	For the correct ratio $OR : RB = 3 : 2$

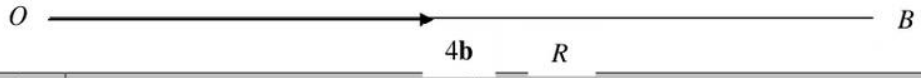
Useful Sketch



Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 11 of 18 · Question 9 part c · 3 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract



Question	Scheme	Mark

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 12 of 18 · Question 10 part a.ii · 3 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 4 · find part a.ii in this mark scheme extract

10(a)	(i) $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB} \Rightarrow \overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$	B1
	(ii) $\overrightarrow{MY} = \overrightarrow{MA} + \frac{3}{4}\left(\overrightarrow{AB}\right) = \mathbf{a} + \frac{3}{4}(-2\mathbf{a} + 4\mathbf{b}) = -\frac{\mathbf{a}}{2} + 3\mathbf{b}$	M1A1 [4]
(b)	$\overrightarrow{OX} = \mu\overrightarrow{OB} = \mu 4\mathbf{b}$	M1
	$\overrightarrow{OX} = \overrightarrow{OM} + \overrightarrow{MX} = \overrightarrow{OM} + \lambda\overrightarrow{MY} = \mathbf{a} + \lambda\left(-\frac{\mathbf{a}}{2} + 3\mathbf{b}\right) = \mathbf{a}\left(1 - \frac{\lambda}{2}\right) + 3\lambda\mathbf{b}$	M1
	$\Rightarrow \mu 4\mathbf{b} = \mathbf{a}\left(1 - \frac{\lambda}{2}\right) + 3\mathbf{b}$	dM1
	$\Rightarrow 1 - \frac{\lambda}{2} = 0 \Rightarrow \lambda = 2$	
	$\Rightarrow 4\mu = 3\lambda \Rightarrow \mu = \frac{6}{4} = \frac{3}{2}$	ddM1
	$OB : OX = 2:3$ oe	A1 [5]
	ALT – working with alternative vector within triangle *OMX*	
	$\overrightarrow{MX} = \overrightarrow{MO} + \overrightarrow{OB} + \mu\overrightarrow{OB} = -\boldsymbol{a} + 4\boldsymbol{b} + \mu 4\boldsymbol{b}$	[M1
	$\overrightarrow{MX} = \lambda\left(-\frac{\boldsymbol{a}}{2} + 3\boldsymbol{b}\right)$	M1
	$\Rightarrow -\boldsymbol{a} = -\frac{\lambda\boldsymbol{a}}{2} \Rightarrow \lambda = 2$	dM1
	$\Rightarrow 4\boldsymbol{b} + \mu 4\boldsymbol{b} = 3\lambda\mathbf{b} \Rightarrow \mu = \frac{1}{2}$	ddM1
	$OB : OX = 2:3$ oe	A1]
(c)	$\frac{\Delta YBX}{\Delta ABX} = \frac{1}{4}$	M1
	$\frac{\Delta ABX}{\Delta OAX} = \frac{1}{3}$	
	$\Rightarrow \frac{\Delta YBX}{\Delta OAX} = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12} \Rightarrow \Delta YBX : \Delta OAX = 1:12$	M1A1 [3]
	ALT – working with relative areas of triangles	
	Area $\Delta YBX = a$ Area $\Delta ABX = 4a$ Area $\Delta OMX = 6a$ Area $\Delta AYX = 3a$ Area $\Delta OAY = 6a$ Area $\Delta OYB = 2a$	[M1
	Area $\Delta OAX = 12a$ $\Delta YBX : \Delta OAX = 1:12$	M1 A1]
		Total 11 marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 12 of 18 · Question 10 part a.ii · 3 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 4 · find part a.ii in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct simplified vector $\overrightarrow{AB}$
	M1	For the correct vector statement for $\overrightarrow{MY}$
	A1	For the correct simplified vector for $\overrightarrow{MY}$
(b)	M1	For the statement $\overrightarrow{OX} = \mu 4\mathbf{b}$ Note: this is a B mark on epen.
	M1	For the correct vector for $\overrightarrow{OX}$ (ft their MY)
	dM1	For equating both vectors for $\overrightarrow{OX}$ and for comparing coefficients of **a** and **b** Dep on M1M1
	ddM1	For finding a value for their parameter for μ Note: there is no mark for only finding λ, they must find μ Dep on M1M1M1
	A1	For the correct ratio $OB:OX = 2:3$ Allow equivalent ratios e.g. 4 : 6, 1 : 1.5
	ALT – working with alternative vector within triangle OMX	
	M1	For a vector statement which includes $\mu 4\boldsymbol{b}$ (for OX or BX) (ft their MY) Note: this is a B mark on epen.
	M1	For a correct second vector equation for the same vector (ft their MY)
	dM1	For equating both vectors and comparing coefficients of **a** and **b** Dep on M1M1
	ddM1	For finding a value for their parameter for μ Note: there is no mark for only finding λ, they must find μ Dep on M1M1M1
	A1	For the correct ratio $OB:OX = 2:3$ Allow equivalent ratios e.g. 4 : 6, 1 : 1.5 Condone $OX:OB = 3:2$ if clearly stated, but not just 3:2
(c)	M1	For either the relationship between the of areas of triangles BYX and ABX **or** the relationship between the areas of triangles ABX and OAX
	M1	For finding the relationship between the of areas of triangles BYX and OAX
	A1	For the correct ratio [1:12] Allow equivalent ratios. Note: do not penalise answer given as a fraction i.e. $\frac{1}{12}$ if already penalised in (b).
	ALT – working with relative areas	
	M1	For assigning a value to one triangle area and writing a second area in terms of this. Note: This could also follow from working with area of a triangle $= \frac{1}{2}ab\sin C$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 12 of 18 · Question 10 part a.ii · 3 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 4 · find part a.ii in this mark scheme extract

		e.g. $\Delta YBX = \frac{1}{2} yz \sin B$ and $\Delta ABX = \frac{1}{2} y(4z) \sin B$
	M1	For finding the relationship between the of areas of triangles BYX and OAX Note: This could also follow from working with area of a triangle $= \frac{1}{2} ab \sin C$ e.g. $\Delta YBX = \frac{1}{2} yz \sin B$ and $\Delta ABX = \frac{1}{2} y(4z) \sin B$
	A1	For the correct ratio [1:12] Allow equivalent ratios. Note: do not penalise answer given as a fraction i.e. $\frac{1}{12}$ if already penalised in (b).

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 12 of 18 · Question 10 part a.ii · 3 marks

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Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 13 of 18 · Question 10 part a.ii · 2 marks

2020 JAN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

10 (a) (i)	$\mathbf{a}+\mathbf{c}$	B1
(ii)	$\frac{1}{2}(\mathbf{c}-\mathbf{a})$	B1 (2)
(b)	$\overrightarrow{OX} = OA + AM + \lambda MN$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda(\frac{1}{2}\mathbf{c}-\frac{1}{2}\mathbf{a})$	M1 A1
	$\mu(\mathbf{a}+\mathbf{c})$	B1
	$\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda\left(\frac{1}{2}\mathbf{c}-\frac{1}{2}\mathbf{a}\right)=\mu(\mathbf{a}+\mathbf{c})$	M1
	$1-\frac{1}{2}\lambda=\mu \quad \frac{1}{2}+\frac{1}{2}\lambda=\mu$	M1
	$1-\frac{1}{2}\lambda=\frac{1}{2}+\frac{1}{2}\lambda$	M1
	$\lambda=\frac{1}{2} \quad \mu=\frac{3}{4}$	A1 A1 (8)
(c)	Triangle $XBN=\frac{1}{8}$ of $\frac{1}{2}$ the parallelogram Quadrilateral $OXNC=\frac{7}{8}$ of $\frac{1}{2}$ the parallelogram So Quadrilateral $OXNC=\frac{7}{16}$ of the parallelogram $\therefore$ 7 : 16	M1 M1 A1 (3) **[13]**

Part	Mark	Additional Guidance
(a)(i)	B1	For the correct vector $\mathbf{a}+\mathbf{c}$
(ii)	B1	For the correct vector $\frac{1}{2}(\mathbf{c}-\mathbf{a})$
(b)	M1	For the correct vector statement $\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AM} + \lambda\overrightarrow{MN}$
	A1	For the correct vector (need not be simplified) $\overrightarrow{OX}=\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$ or $\overrightarrow{OX}=\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda\left(\frac{\mathbf{c}}{2}-\frac{\mathbf{a}}{2}\right)$
	B1ft	For $\overrightarrow{OX}=\mu(\mathbf{a}+\mathbf{c})$ ft their $\overrightarrow{OB}=$'$\mathbf{a}+\mathbf{c}$'
	M1	For equating their two vector statements for $\overrightarrow{OX}$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})=\mu(\mathbf{a}+\mathbf{c})$
	M1	For equating coefficients of $\mathbf{a}$ and $\mathbf{c}$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})=\mu(\mathbf{a}-\mathbf{c})\Rightarrow\mathbf{a}\left(1-\frac{\lambda}{2}\right)+\mathbf{c}\left(\frac{1}{2}+\frac{\lambda}{2}\right)=\mu\mathbf{a}+\mu\mathbf{c}$ $\Rightarrow\mu=1-\frac{\lambda}{2},\ \mu=\frac{1}{2}+\frac{\lambda}{2}$
	M1	For attempting to solve their two simultaneous equations in terms of λ and μ. $1-\frac{\lambda}{2}=\frac{1}{2}+\frac{\lambda}{2}\Rightarrow\lambda=\ldots\Rightarrow\mu=\ldots$ or $1-\mu=\mu-\frac{1}{2}\Rightarrow\mu=\ldots\Rightarrow\lambda=\ldots$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 13 of 18 · Question 10 part a.ii · 2 marks

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	A1	For **either** $\lambda = \frac{1}{2}$ **or** $\mu = \frac{3}{4}$
	A1	For **both** $\lambda = \frac{1}{2}$ **and** $\mu = \frac{3}{4}$
	ALT	
	M1	For the correct vector statement $\overrightarrow{MX} = \overrightarrow{MO} + \overrightarrow{OX}$
	A1	For the correct vector (need not be simplified) $\overrightarrow{MX} = -\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c})$
	B1ft	$\overrightarrow{MX} = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$ ft their $\overrightarrow{MN} = '\frac{1}{2}(\mathbf{c}-\mathbf{a})'$
	M1	For equating the two vector statements for $\overrightarrow{MX}$ $-\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c}) = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$
	M1	For equating coefficients of $\mathbf{a}$ and $\mathbf{c}$ $-\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c}) = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a}) \Rightarrow \mathbf{c}\left(-\frac{1}{2}+\mu\right) + \mathbf{a}(\mu-1) = \mathbf{c}\frac{\lambda}{2} - \mathbf{a}\frac{\lambda}{2}$ $\Rightarrow \frac{\lambda}{2} = \mu - \frac{1}{2}$ and $-\frac{\lambda}{2} = \mu - 1$
	M1	For attempting to solve their two simultaneous equations in terms of λ and μ. $\mu - \frac{1}{2} = -(\mu-1) \Rightarrow \mu = \left(\frac{3}{4}\right)$ $\frac{\lambda}{2} = 1 - '\frac{3}{4}' \Rightarrow \lambda = \left(\frac{1}{2}\right)$
	A1	For **either** $\lambda = \frac{1}{2}$ **or** $\mu = \frac{3}{4}$
	A1	For **both** $\lambda = \frac{1}{2}$ **and** $\mu = \frac{3}{4}$
(c)	M1	For area of $\Delta XBN = \frac{1}{8}\Delta OBC$ so $\frac{1}{8}$ of $\frac{1}{2}$ of the area of parallelogram $OABC$ $\left(\Delta OBC = \frac{1}{2} \times OB \times BC \times \sin\angle XBN \right.$ $\left. \Delta XBN = \frac{1}{2} \times \frac{1}{4} OB \times \frac{1}{2} BC \times \sin\angle XBN = \frac{1}{8}\Delta OBC \right)$
	M1	Therefore Quadrilateral $OXNC = \frac{7}{8}$ of $\frac{1}{2}$ of the area of parallelogram $OABC$ ft their fraction from the first M mark provided it is $< \frac{1}{2}$
	A1	Quadrilateral $OXNC = \frac{7}{16}$ of the area of parallelogram $OABC$ so ratio is 7:16

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 13 of 18 · Question 10 part a.ii · 2 marks

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Question number	Scheme	Marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 14 of 18 · Question 10 part a.i · 4 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

10 a (i)	$\overrightarrow{OP} = \overrightarrow{OA} + \frac{3}{4}\overrightarrow{AB} = \mathbf{a} + \frac{3}{4}(-\mathbf{a}+\mathbf{b})$ or $\overrightarrow{OB} + \frac{1}{4}\overrightarrow{BA} = \mathbf{b} + \frac{1}{4}(\mathbf{a}-\mathbf{b})$	M1
	$\overrightarrow{OP} = \frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$	A1
(ii)	$\overrightarrow{MN} = \overrightarrow{MO} + \frac{1}{2}\overrightarrow{OP} = -\frac{1}{2}\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$ or $\frac{1}{2}\overrightarrow{AO} + \frac{1}{2}\overrightarrow{OP}$	M1
	$\overrightarrow{MN} = -\frac{3}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$	A1 (4)
b	$\overrightarrow{OC} = \lambda\mathbf{b}$	B1
	$\overrightarrow{AN} = \left(\overrightarrow{AO} + \overrightarrow{ON}\right) = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right) = -\frac{7}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$ or $\overrightarrow{AN} = \overrightarrow{AP} + \overrightarrow{PN} =$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \mu\overrightarrow{AN} = \mathbf{a} + \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$	M1 (A1 on ePen)
	$= \mathbf{a} - \frac{7}{8}\mu\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$ *or* $\left(1 - \frac{7}{8}\mu\right)\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$	A1 (M1 on ePen)
	$\therefore \lambda = \frac{3}{8}\mu$ and $0 = 1 - \frac{7}{8}\mu \Rightarrow \mu = \frac{8}{7}$ $\therefore \lambda = \frac{3}{7}$	M1 (A1 on ePen)
	$\overrightarrow{OC} = \frac{3}{7}\mathbf{b}$	A1 (6)
ALT	$\overrightarrow{AC} = -\mathbf{a} + \lambda\mathbf{b}$	B1
	$\overrightarrow{AN} = \left(\overrightarrow{AO} + \overrightarrow{ON}\right) = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right) = -\frac{7}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$ or $\overrightarrow{AN} = \overrightarrow{AP} + \overrightarrow{PN} =$	M1
	$\overrightarrow{AC} = \mu\overrightarrow{AN} = \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$	M1 (A1 on ePen)
	$= -\frac{7}{8}\mu\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$	A1 (M1 on ePen)
	$\therefore \lambda = \frac{3}{8}\mu$ and $0 = 1 - \frac{7}{8}\mu \Rightarrow \mu = \frac{8}{7}$ $\therefore \lambda = \frac{3}{7}$	M1 (A1 on ePen)
	$\overrightarrow{OC} = \frac{3}{7}\mathbf{b}$	A1 (6)

General principles for marking part (b) – if in any doubt about allocating marks – send to review

B1 Writes a valid vector with a parameter in terms of **a** or **b** which leads to finding $\overrightarrow{OC}$

M1 M1 A1 Writes a second valid vector with a different parameter, in terms of **a** and **b**, following a distinct different route, which leads to finding $\overrightarrow{OC}$

M1 Compares components with two different parameters and arrives at a value for μ or λ

A1 correct vector

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 14 of 18 · Question 10 part a.i · 4 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

c	(Area triangle) $OAP = \frac{3}{4}$ (Area triangle) OAB	B1
	(Area triangle) $OMN = \frac{1}{4}$ (Area triangle) $OAP = \frac{3}{16}$ (Area triangle) OAB	B1
	(Area quadrilateral) $AMNP =$ $\frac{3}{4}$ (Area triangle) $OAB - \frac{3}{16}$ (Area triangle) OAB	M1
	$= \frac{9}{16}$ (Area triangle) OAB $\quad k = \frac{9}{16}$	A1 (4)
ALT	$\frac{(\text{Area triangle})OAP}{(\text{Area triangle})OAB} = \frac{3}{4}$	B1
	$\left(\frac{\text{Area triangle } OMN}{\text{Area triangle } OAP} = \frac{1}{4}\right) \Rightarrow \frac{(\text{Area quadrilateral})MNAP}{(\text{Area triangle})OAP} = \frac{3}{4}$	B1
	$\frac{(\text{Area quadrilateral})MNAP}{(\text{Area triangle})OAP} \times \frac{(\text{Area triangle})OAP}{(\text{Area triangle})OAB} = \frac{3}{4} \times \frac{3}{4}$	M1
	$k = \frac{9}{16}$	A1 (4)
		Total 14 marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 14 of 18 · Question 10 part a.i · 4 marks

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Part	Marks	Note
(a) (i)	M1	For stating **or** using $\overrightarrow{OP} = \overrightarrow{OA} + \frac{3}{4}\overrightarrow{AB}$ or for $\mathbf{a} + \frac{3}{4}(-\mathbf{a} + \mathbf{b})$ or $\overrightarrow{OB} + \frac{1}{4}\overrightarrow{BA}$ or $\mathbf{b} + \frac{1}{4}(\mathbf{a} - \mathbf{b})$ (can be implied by correct vector)
	A1	For $\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ or valid alternative form such as $\frac{\mathbf{a}+3\mathbf{b}}{4}$ or $\frac{1}{4}(\mathbf{a}+3\mathbf{b})$
(ii)	M1	For stating **or** using $\overrightarrow{MN} = \overrightarrow{MO} + \frac{1}{2}\overrightarrow{OP}$ or $\frac{1}{2}\overrightarrow{AO} + \frac{1}{2}\overrightarrow{OP}$ or $-\frac{1}{2}\mathbf{a} + \frac{1}{2}\left(\text{their } \overrightarrow{OP}\right)$
	A1	$-\frac{3}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$ or valid alternative form such as $\frac{-3\mathbf{a}+3\mathbf{b}}{8}$ or $\frac{1}{8}(-3\mathbf{a}+3\mathbf{b})$ (can be implied by correct vector)
(b)	B1	$\overrightarrow{OC} = \lambda\mathbf{b}$ or any equivalent statement involving a parameter, in terms of **b**.
	M1	A fully correct method to find $\overrightarrow{AN}$ using their $\overrightarrow{OP}$ $\overrightarrow{AN} = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$
	M1 (A1 on ePen)	**Using** $\overrightarrow{OC} = \overrightarrow{OA} + \mu\left(\text{their } \overrightarrow{AN}\right)$ in terms **a** and **b** $\overrightarrow{OC} = \overrightarrow{OA} + \mu\overrightarrow{AN} = \mathbf{a} + \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$ Simplification not required
	A1 (M1 on ePen)	$= \mathbf{a} - \frac{7}{8}\mu\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$ *or* $\left(1 - \frac{7}{8}\mu\right)\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$ either of these forms, ready for comparing coefficients
	M1 (A1 on ePen)	For correctly equating their components with two different parameters and attempting to solve, reaching values for μ or λ We need to see two equations here, leading to a value for one of the parameters.
	A1	$\overrightarrow{OC} = \frac{3}{7}\mathbf{b}$
ALT	B1	$\overrightarrow{AC} = -\mathbf{a} + \lambda\mathbf{b}$ or any equivalent statement involving a parameter, in terms of **a** and **b**.
	M1	A fully correct method to find $\overrightarrow{AN}$ using their $\overrightarrow{OP}$ $\overrightarrow{AN} = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$
	M1 (A1 on ePen)	$\overrightarrow{AC} = \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$ using their $\overrightarrow{AN}$
	A1 (M1 on ePen)	Correct vector for $\overrightarrow{AC}$
	M1 (A1 on ePen) A1	Marks allocated as main scheme
(c) BOTH SCHEMES	B1	Correct statement
	B1	Correct statement
	M1	Uses their statements to carry out a relevant calculation
	A1	Correct value for k

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 14 of 18 · Question 10 part a.i · 4 marks

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Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 15 of 18 · Question 8 part c · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

8 (a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \mathbf{a} + \mathbf{b}$	B1 [1]
(b)	$\overrightarrow{OC} = 2\mathbf{b}$ $\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC} = -(\mathbf{a} + \mathbf{b}) + 2\mathbf{b} = \mathbf{b} - \mathbf{a}$	B1 M1A1 [3]
(c)	$\overrightarrow{OM} = \overrightarrow{OB} + \overrightarrow{BM} = \mathbf{a} + \mathbf{b} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}$ or $\frac{1}{3}(\mathbf{a} + 5\mathbf{b})$	 M1A1ft [2]
(d)	$\overrightarrow{OY} = \mu\left(\frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}\right) = \frac{\mu\mathbf{a}}{3} + \frac{5\mu\mathbf{b}}{3}$ $\overrightarrow{OY} = \overrightarrow{OA} + \overrightarrow{AY} = \boldsymbol{a} + \lambda\boldsymbol{b}$ $\Rightarrow \frac{\mu}{3} = 1$ and $\frac{5\mu}{3} = \lambda$ Solves simultaneous equations by any method $\mu = 3,\ \lambda = 5$ $AB : BY = 1 : 4$ **ALT** $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ $\Rightarrow -1 + \frac{1}{3}\mu = 0$ and $\lambda = -1 + \frac{5}{3}\mu$ $\mu = 3, \lambda = 4$ $AB : BY = 1 : 4$	M1 M1 M1 M1 A1 [5] [M1 M1 M1 M1 A1]
(e)	$10 = \frac{1}{2}ab\sin 60^\circ \Rightarrow ab = \frac{40}{\sqrt{3}} \Rightarrow a = \frac{40}{b\sqrt{3}}$ $\text{Area} = \frac{1}{2}a \times 5b\sin 120^\circ = \frac{1}{2} \times \frac{40}{b\sqrt{3}} \times 5b\sin 120^\circ$ $\text{Area} = 50$ **ALT** $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2} \times h \times 5}{\frac{1}{2} \times h \times 1} = \frac{5}{1}$ $\text{Area } OAY = 5 \times \text{Area } OAB$ $\text{Area } OAB = 10$ $\text{Area} = 50$	M1A1 dM1 A1 [4] [M1 A1 dM1 A1]
		Total 15 marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 15 of 18 · Question 8 part c · 2 marks

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Part	Mark	Notes
(a)	B1	For a correct expression for $\overrightarrow{OB}$ in terms of **a** and **b**
(b)	B1	For a correct expression for $\overrightarrow{OC}$
	M1	For a correct vector statement for $\overrightarrow{BC}$: $\overrightarrow{BC}=\overrightarrow{BO}+\overrightarrow{OC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$. Vector statement must be suitable for substitution to find $\overrightarrow{BC}$
	A1	For the correct simplified $\overrightarrow{BC}$ in terms of a single **a** and **b** only. $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ If answer $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ seen without wrong working then award B1M1A1.
(c)	M1	For a correct vector statement for $\overrightarrow{OM}$: $\overrightarrow{OM}=\overrightarrow{OB}+\frac{2}{3}\overrightarrow{BC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$ and their $\overrightarrow{BC}$.
	A1ft	For the correct simplified using their $\overrightarrow{OM}$ in terms of a single **a** and **b** only. $\overrightarrow{OM} = \frac{\boldsymbol{a}}{3} + \frac{5\boldsymbol{b}}{3}$ or $\frac{1}{3}(\boldsymbol{a} + 5\boldsymbol{b})$
(d)	M1 M1	M1 for one correct statement of route for $\overrightarrow{OY}$ M1 for second correct statement of route for $\overrightarrow{OY}$ $\overrightarrow{OY} = \mu\overrightarrow{OM}$ (or any other variable in place of μ) $\overrightarrow{OY} = \left(\overrightarrow{OA} + \overrightarrow{AY}\right) = \mathbf{a} + \lambda\mathbf{b}$ (or any other variable in place of λ, provided this is different to their μ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$
ALT – use of $\overrightarrow{BY}$ route, use also for $\overrightarrow{AY}$ route		
	M1 M1	M1 for one correct statement of route for $\overrightarrow{BY}$ M1 for second correct statement of route for $\overrightarrow{BY}$ $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ (or any other variable in place of λ) $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ (or any other variable in place of μ provided this is different to their λ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 15 of 18 · Question 8 part c · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

(e)	**M1**	For use of the **correct** formula for area of a triangle with $60°$ and correct value of $\sin 60°$ $10 = \frac{1}{2}ab\sin 60° = ab \times \frac{\sqrt{3}}{4}$
	A1	For correct expression for ab or a $ab = \frac{40}{\sqrt{3}}$ or $a = \frac{40}{b\sqrt{3}}$
	dM1	For use of the **correct** formula for area of a triangle with 120° and attempt to substitute for ab or a. $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times '\frac{40}{b\sqrt{3}}' \times '5'b\sin 120°$ or $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times ab \times '5'\sin 120° = \frac{1}{2} \times '\frac{40}{\sqrt{3}}' \times '5'\sin 120°$ Dependent on the first M awarded. Allow use of their λ from part (d).
	A1	For the correct area $\text{Area} = 50$
ALT – use of ratios of areas		
	M1	For use of their ratio $AB:BY$ to write an equation linking area OAY and area OAB $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2}\times h\times(1+'4')}{\frac{1}{2}\times h\times 1} = \frac{'5'}{1}$
	A1	For correct relationship between area OAY and area OAB
	dM1	For a correct method to find the area of OAB Dependent on first M mark being awarded.
	A1	For the correct area $\text{Area} = 50$

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 15 of 18 · Question 8 part c · 2 marks

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Question number	Scheme	Marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 16 of 18 · Question 11 part a · 1 mark

2020 NOV · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Question number		
11 (a)	$\mathbf{b}-\mathbf{a}$	B1 (1)
(b)	$\overrightarrow{OZ}=\overrightarrow{OB}+\lambda\overrightarrow{BX}(=\mathbf{b}+\lambda(-\mathbf{b}+2\mathbf{a}))$	M1
	$=(1-\lambda)\mathbf{b}+2\lambda\mathbf{a}$	A1
	$\overrightarrow{OZ}=\overrightarrow{OA}+\mu\overrightarrow{AY}(=\mathbf{a}+\mu(-\mathbf{a}+3\mathbf{b}))$	M1
	$=(1-\mu)\mathbf{a}+3\mu\mathbf{b}$	A1
	$(1-\lambda)\mathbf{b}+2\lambda\mathbf{a}=(1-\mu)\mathbf{a}+3\mu\mathbf{b}$	ddM1
	$2\lambda=1-\mu$ $3\mu=1-\lambda$	A1
	$3(1-2\lambda)=1-\lambda$ or $2(1-3\mu)=1-\mu$	M1
	$\lambda=\frac{2}{5}$ or $\mu=\frac{1}{5}$	A1
	$\overrightarrow{OZ}=\frac{1}{5}(4\mathbf{a}+3\mathbf{b})$ See notes regarding alternatives	A1 (9)
(c)	$\overrightarrow{OM}=p"\frac{1}{5}(4\mathbf{a}+3\mathbf{b})"$ and $\overrightarrow{OM}=2\mathbf{a}+q(-2\mathbf{a}+3\mathbf{b})$	M1
	$\frac{4p}{5}=2-2q$ and $\frac{3p}{5}=3q$ (Solving these equations leads to $p=\frac{5}{3}$)	M1
	$\overrightarrow{OM}=\frac{1}{3}(4\mathbf{a}+3\mathbf{b})$	A1 (3)
		[13]

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 16 of 18 · Question 11 part a · 1 mark

2020 NOV · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For the indicated vector
(b)	M1	For any correctly written vector path, must include a parameter
	A1	For the vector shown
	M1	For any correctly written vector path, must include a parameter
	A1	For the vector shown
	ddM1	Equates their 2 vectors – this mark may be implicit in the candidate equating the two components of their 2 vectors, **dependent on the first two method marks.**
	A1	Correct equations as shown
	ddM1	Full and correct method to solve their two simultaneous equations, either by substitution as shown or by elimination. There must be no errors in the method to eliminate λ or μ, **dependent on the first two method marks.**
	A1	Correct value for λ or μ
	A1	Correct vector.
	There are a number of alternatives for part b, all marked in the same way. Examples: ALT1 $\mu\overrightarrow{XB}$ and $\lambda\overrightarrow{AY}$ (must use to get $\overrightarrow{AB}$ for 2nd A1) equate 2 vectors for $\overrightarrow{AB}$ ALT2 $\overrightarrow{AZ} = \overrightarrow{AX} + \mu\overrightarrow{XB}$ M1A1 $\overrightarrow{AZ} = \lambda\overrightarrow{AY}$ M1A1 equate 2 vectors for $\overrightarrow{AZ}$ ddM1 The A1 ddM1 A1 A1 following these marks should all be marked in the same way as the main mark scheme.	
(c)	M1	For the two correct vectors shown, allow use of their $\overrightarrow{OZ}$
	dM1	Correctly equating the components of their vectors for $\overrightarrow{OZ}$ and arriving at a value for p or q
	A1	For the correct vector, as shown.
	Can also be done using other vectors eg finding two alternatives for $\overrightarrow{OM}$ Mark in the same way as main scheme.	

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 16 of 18 · Question 11 part a · 1 mark

2020 NOV · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 17 of 18 · Question 6 part c · 2 marks

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6 (a)	eg $\overrightarrow{SR} = \overrightarrow{SP} + \overrightarrow{PR} = (3\mathbf{i} - 15\mathbf{j} - \mathbf{i} + 18\mathbf{j})$ $= 2\mathbf{i} + 3\mathbf{j}$ or eg $\overrightarrow{QR} = \overrightarrow{QP} + \overrightarrow{QR} = (-2\mathbf{i} - 3\mathbf{j} - \mathbf{i} + 18\mathbf{j})$ $= -3\mathbf{i} + 15\mathbf{j}$	M1 A1 M1 A1
	eg $\overrightarrow{PQ} = \overrightarrow{SR}$ or $\overrightarrow{PS} = \overrightarrow{QR}$ oe	M1
	(As opposite sides are equal in length and parallel) $PQRS$ is a parallelogram *	A1cso [4]
(b)	$\overrightarrow{QS} = \overrightarrow{QP} + \overrightarrow{PS} = (-2\mathbf{i} - 3\mathbf{j} - 3\mathbf{i} + 15\mathbf{j})$ $= -5\mathbf{i} + 12\mathbf{j}$	M1 A1
	Unit vector = $(\pm)\frac{1}{13}(-5\mathbf{i} + 12\mathbf{j})$	M1 A1 [4]
(c)	$\overrightarrow{PT} = 2\mathbf{i} + 3\mathbf{j} + \frac{5}{13}("-5\mathbf{i} + 12\mathbf{j}") = \frac{1}{13}\mathbf{i} + \frac{99}{13}\mathbf{j}$	M1 A1 [2]
		Total 10 marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 17 of 18 · Question 6 part c · 2 marks

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Question number	Scheme	Marks

Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 18 of 18 · Question 6 part c · 3 marks

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number		
6 (a)	$\left(\overrightarrow{AB}\right) = \overrightarrow{AO} + \overrightarrow{OB} = -9\mathbf{a} - 12\mathbf{b} + 2\mathbf{a} + p\mathbf{b}$	M1
	$(-7)^2 + (p-12)^2 = 25^2$	M1
	$(p-12)^2 = 576$	
	$p = 12 \pm \sqrt{576}$	M1
	$p = -12, 36^*$	A1*
		[4]
(b)	$\sqrt{12^2 + 2^2} \Rightarrow \left(\sqrt{148} = 2\sqrt{37}\right)$	M1
	$\pm\frac{1}{\sqrt{37}}(\mathbf{a} - 6\mathbf{b})$ **or** $\pm\frac{\sqrt{37}}{37}(\mathbf{a} - 6\mathbf{b})$	A1
		[2]
(c)	$(\overrightarrow{OC}) = \frac{2}{5}(2\mathbf{a} + 36\mathbf{b})$	M1
	$(\overrightarrow{CA}) = -\frac{2}{5}(2\mathbf{a} + 36\mathbf{b}) + 9\mathbf{a} + 12\mathbf{b}$	M1
	$= \frac{41}{5}\mathbf{a} - \frac{12}{5}\mathbf{b}$	A1
		[3]
		Total 9 marks

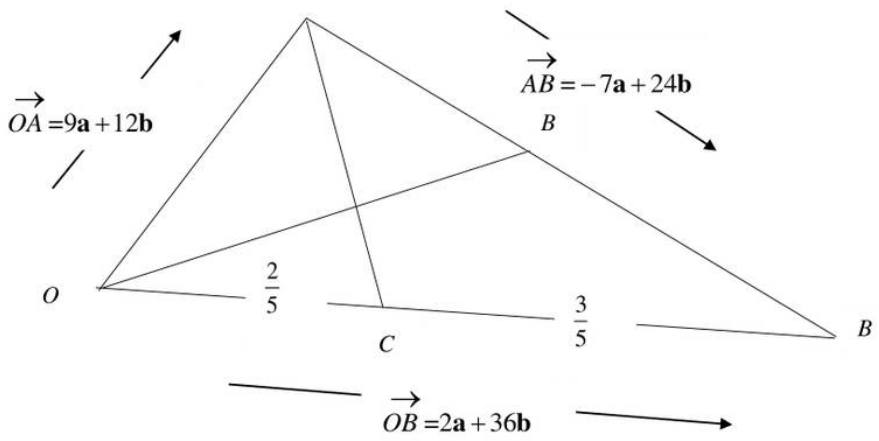
Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 18 of 18 · Question 6 part c · 3 marks

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Part	Mark	Additional Guidance
6 (a)	M1	For a correct vector path to find $\pm\overrightarrow{AB}$ or $-9\mathbf{a} - 12\mathbf{b} + 2\mathbf{a} + p\mathbf{b}$ or $9\mathbf{a} + 12\mathbf{b} - 2\mathbf{a} - p\mathbf{b}$
	M1	For using Pythagoras' Theorem with th B $\overrightarrow{AB}$
	M1	For any complete and minimally acceptable attempt to solve for p See General Guidance.
	A1*	For both correct values of p – no errors seen. **Note:** This is a show question and the answers are given
(b)	M1	For a corr od to find the magnitude.
O	A1	For a corr $\frac{2}{5}$ vector as shown.
(c)	M1	For a corr or fc $\overrightarrow{C}$
	M1	For the correct vector for $\overrightarrow{CA}$ using their $\overrightarrow{CO} + \overrightarrow{OA}$
	A1	For $\frac{41}{5}\mathbf{a} - \frac{12}{5}\mathbf{b}$
	ALT: Uses the vector path $\overrightarrow{CA} = \overrightarrow{CB} + \overrightarrow{BA}$	
	M1	For a correct vector for $\overrightarrow{CB} = \frac{3}{5}(2\mathbf{a} + 36\mathbf{b})$ or $\overrightarrow{BA} = 7\mathbf{a} - 24\mathbf{b}$
	M1	For the correct vector for $\overrightarrow{CA}$ using their $\overrightarrow{CB}$ and $\overrightarrow{BA}$ $\overrightarrow{CA} = \frac{3}{5}(2\mathbf{a} + 36\mathbf{b}) + (7\mathbf{a} - 24\mathbf{b})$
	A1	$\overrightarrow{CA} = \frac{41}{5}\mathbf{a} - \frac{12}{5}\mathbf{b}$

Useful Sketch



Type MS · Finding a route: vectors round a shape with midpoints and ratios

Answer 18 of 18 · Question 6 part c · 3 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks